

STEM4U — LEGO Computer Science & AI · Grades K-2 (Sparks) · 30-Lesson Plan

Course Overview

Thirty weekly 60-minute sessions built on the LEGO® Education Computer Science & AI Kit (45520), coded with **icon-only picture blocks** on LEGO's Coding Canvas — no reading or typing required, and a simple PIN gets kids in (no logins, no accounts). Groups of 4 work in pairs (Driver/Navigator). Single grade band, grades K-2 (the **Sparks**). Real tools. Real teams. Real engineers.

By the end, your child can build and code a small machine that runs steps in order, learns what a motor and a color sensor do, reacts when something happens, repeats a pattern, and collects data — and they can explain, in their own words, how a computer learns to recognize things from examples.

AI runs through every unit — honestly. Kids don't wait until Lesson 26 to meet artificial intelligence. Every lesson includes a 2-minute "AI Connection" (the **coach reads it aloud**) that ties the day's coding idea to how real AI works. At this age the promise is clear and true: **K-2 kids explore how AI works — they do not build or train their own AI model.** In the final unit they collect and show simple data, then **use ready-made AI pose-recognition blocks** that already know how to spot body poses, plus a short coach-run demo (Google Teachable Machine) that learns from examples the kids feed it. The learning belongs to the ready-made blocks and the coach's demo, not the child. That honest framing is a feature, not a limitation — it leaves the real "train your own classifier" work for the 3-5 band, where it belongs.

Scope & sequence

Unit	Lessons	Big idea	What your child can do by unit's end
1 · Basics	1-5	A program is steps in order	Plan and code a machine that runs steps they put in order
2 · Hardware	6-10	A machine's parts follow our code	Start and stop a motor and make it work with a color sensor
3 · Events	11-15	Something makes the program start	Make a model react to a press, a color, or a signal
4 · Sequences	16-20	The right order gets the job done	Build, fix, and tell a story with an ordered program
5 · Loops	21-25	Computers can repeat for us	Build a machine that repeats a pattern and keeps count
6 · AI & Data	26-30	Machines sort, show data, and recognize	Collect data and use ready-made AI pose blocks to drive a build

Standards & bridges: Lessons are aligned to the CSTA K-12 CS Standards — primarily Algorithms & Programming (1A-AP), with connections to Data & Analysis (1A-DA) and Impacts of Computing (1A-IC). (See the **Standards Crosswalk** at the end for the specific codes per unit.) The lessons build the earliest mental models — order, how the parts work, triggers, getting the sequence right, repeating, and collecting data — that bridge toward the 3-5 band and, later, ScratchJr and Scratch.

Rotate the "Concept + demo" format so the same 8 minutes feels fresh week to week. Pick one each lesson: **(a) straight demo** (coach shows the idea), **(b) "the silly robot"** (coach codes it wrong on purpose; kids catch the mistake and cheer the fix), **(c) "guess what it does"** (predict with a thumbs-up/down before the coach runs it), or **(d) "show a friend"** (a pair who gets it shows another pair). Keep demos short — little ones watch best in 2-minute bursts, so talk, run, react, repeat.

Every week gets a build/ownership hit. On mostly-code weeks, spend a spare minute on a "**make-it-yours**" **beat** — snap on one brick, give the bot a LEGO face, or name it — so no week is screen-only. Little hands need to touch bricks every session.

The six design challenges are themed so each feels like a party, with a varied show-and-tell: **L5 Build-a-Bot parade** (bots march on their trays, gallery walk) · **L10 Signal fair** (everyone shows their color signal machine) · **L15 Gadget fair** ("make my model go!") · **L20 Story theater** (pairs perform their in-order story) · **L25 Pattern party** (everyone's machine repeats at once, like a band) · **L30 AI pose show** (perform for a short parent-note video).

Progress layer: Each unit ends with a design-challenge lesson (5, 10, 15, 20, 25, 30). At each of these, the coach sends home a **short parent note** ("here's what your child can do now") plus a **photo of the design-challenge build**, so families can see growth across the year.

How to read a lesson (for coaches with no CS background)

Every lesson follows the same shape and the same clock, so you can teach it cold: an "I can..." goal, a 2-line AI Connection to **read aloud**, a materials note (most weeks the model is already built from earlier in the unit), a **timed 60-minute flow**, Driver/Navigator pairing, differentiation (Support / Extension / Reteach), an exit check, and an optional short at-home idea. Extensions double as **fast-finisher** tasks; the Reteach line is your quick fix if a pair is stuck. Keep your own instructions to one short sentence at a time — model it, don't explain it.

The 60-minute clock (every lesson): Warm-up (song/story/movement) **10** · Concept + demo **8** · Build/Code **25** (with a stand-up wiggle break at the halfway swap) · Challenge **7** · Share + tidy-up **10**. The clock gives little kids more movement at the start and a longer, calmer pack-up — younger focus comes in short bursts, so we front-load energy and keep hands-on time in a 25-minute window.

Model reuse: Each unit's LEGO model is built once, in the unit's first lesson, and stays assembled for the rest of the unit. Later lessons in a unit are mostly coding (with only small changes), so class time goes to thinking and playing, not rebuilding. Store each group's model on a labeled tray between sessions. The kit uses **one motor and one color sensor** — the two parts kids get to know by name all year.

Pairing: Driver taps the picture blocks and presses Run; Navigator holds the picture-card plan and points to the next step. They **swap at the halfway mark** of Build/Code every lesson (that's the wiggle break — stand, stretch, switch seats). Odd number of kids? Make one **rotating trio** — three jobs (Driver, Navigator, Tester) that rotate at each third. Because readers and pre-readers mix at this age, pair a confident picture-reader with a partner when you can.

Lesson 1 — Basics: Meet the Kit & First Press

Objective (I can...): I can find my way around the Coding Canvas with pictures, work with a partner, and press two picture blocks together to make a motor move. **CS concept:** What a program is (steps a machine follows) · **Builds on:** No prior lesson — course launch.

AI Connection (2 min, read aloud): A tablet that unlocks when it sees your face is a computer following steps and making a choice. *Ask the group:* "If a robot wanted to know 'is this a cat?', what tiny things could it look at?" (Just collect ideas — we'll come back to this all year.)

Materials: LEGO CS & AI kit 45520 (1 per pair), a device per pair (Coding Canvas, PIN ready — no login), a big printed "kit parts" picture card per pair, a "team jobs" card (Driver / Navigator pictures). **No full build today** — use the kit's quick-start model (hub plus one motor already on). The unit's real model is built next lesson.

60-minute flow - Warm-up (10): "Robot chef" — a child gives the coach spoken steps to make a pretend sandwich; the coach does exactly what's said, silly mistakes and all (butters the plate, etc.). Big point, lots of giggles: machines do EXACTLY what you tell them. Then teach the two team jobs with a quick chant: "Driver drives, Navigator points." - **Concept + demo (8):** Turn on the hub; show it connecting to the tablet. Point to the picture palette, the big green Run, and the red Stop. Drag a start picture + one "spin motor" picture; press Run so a real motor moves. Let them gasp. - **Build/Code (25):** Each pair (1) gets their hub connected with coach help, (2) takes a picture tour — find the palette, Run, Stop, (3) places 2 picture blocks (start + spin) on the quick-start model and presses Run. *Wiggle break + swap at ~12 min, and practice trading the two jobs.* Coach circulates so every hub is connected. - **Challenge (7):** Add a "stop after a moment" picture block so the motor stops on its own. Change which picture is first and guess what happens. - **Share + tidy-up (10):** Each pair presses Run for the group and points to one kit part by its picture. Coil cables, hub off, parts in the tray. Sing a "clean-up" count-down.

Pair roles: Driver connects (with help) and drags pictures; Navigator holds the parts card and finds each part. Swap at the wiggle break. *(Trio: add a Tester who presses Run and tells the group what moved.)*

Differentiation: *Support* — a palette card so kids match blocks by picture, not words. *Extension / fast-finisher* — add a sound picture when the motor stops. *Reteach* — connect the hub together slowly; have the child name each button before pressing.

Exit check: Pair has a working hub, ran a 2-block program that moves a motor, can point to the hub, a motor, and the green Run, and can name both team jobs. **At-home (optional):** Tell a grown-up 4 exact steps to put on your shoes, like talking to a robot. ~5 min.

Lesson 2 — Basics: Everyday Steps in Order

Objective (I can...): I can build the unit model and put picture blocks in order so it does one thing after another, like the steps in an everyday job. **CS concept:** Algorithms / sequencing (everyday jobs are steps in order; steps run top to bottom) · **Builds on:** Lesson 1 (running a program).

AI Connection (2 min, read aloud): When you talk to a voice helper, it follows steps in order: hear you → think → answer. *Ask the group:* "What if it tried to answer BEFORE it heard you? Would that work?"

Materials: LEGO CS & AI kit 45520 (1 per pair), device per pair, 4 big "make toast" picture cards per pair.

This is the unit's build lesson — build the Basics model now and keep it assembled through Lesson 5.

60-minute flow - Warm-up (10): Card scramble — pairs put 4 out-of-order "make toast" picture cards in the right order on the rug, then act it out. Laugh at what happens if you eat the toast before you toast it. Name it: an everyday job is a list of steps in order — that's an algorithm. - **Concept + demo (8):** Show the same picture blocks in two different orders; run both so kids SEE different results. Say the rule together: "top to bottom, in order." - **Build/Code (25):** Pairs **build the Basics model** following the Canvas picture steps (the one build for the unit), then make a 3–4 step program (light on → motor spins → wait → stop) and run it, pointing at each step as it happens. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Move the same blocks into a new order that still makes sense; run and say what changed. - **Share + tidy-up (10):** Each pair shows "order 1" vs. "order 2." **Leave the model built** on its tray.

Pair roles: During the build, Driver snaps bricks while Navigator points to the picture steps; during coding they swap. Swap at the wiggle break. (*Trio rotates build/point/test.*)

Differentiation: *Support* — pre-place the start block so they add the rest. *Extension / fast-finisher* — make a 5-step program. *Reteach* — touch each block and count "1, 2, 3, 4" out loud, running one step at a time.

Exit check: Pair runs an ordered program and can show why a different order gave a different result. **At-home (optional):** Draw a 4-picture comic of waking up, in order. ~5 min.

Lesson 3 — Basics: Break the Big Job into Little Steps

Objective (I can...): I can take one big action and break it into little steps before I code it. **CS concept:** Decomposition (splitting a job into smaller parts) · **Builds on:** Lesson 2 (ordered steps).

AI Connection (2 min, read aloud): To spot a cat, a computer looks at little clues — pointy ears, whiskers, a tail — then puts them together. *Ask the group:* "What little clues tell YOU it's a cat and not a dog?"

Materials: LEGO CS & AI kit + device per pair, big sticky notes or a mini picture-plan mat per pair. **Model reuse:** Basics model from Lesson 2 (no rebuild) — code plus one tiny brick tweak.

60-minute flow - Warm-up (10): "Program the human robot" — one child is the robot; the partner gives only tiny steps ("step... step... turn") to walk them to a spot. If a step is too big or fuzzy, the robot freezes with arms out. Point: machines need little, exact steps — that's breaking it down. - **Concept + demo (8):** Coach picks "make it wave hello," says the little steps out loud, then turns each into one picture block. (*Try "the silly robot" — leave out a step and let them catch it.*) - **Build/Code (25):** Using the built model, pairs put one picture-step per sticky note first, then place one block per step and run. **Make-it-yours beat:** add one brick or a LEGO face and name the bot. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add one more little step (a light or a sound) and show where it goes in the line. - **Share + tidy-up (10):** Pairs show their sticky-note line next to the running, now-named bot. Model stays built.

Pair roles: Navigator lays out the sticky-note steps; Driver codes them. Swap at the wiggle break. (*Trio: third child points to each step as it runs.*)

Differentiation: *Support* — a half-filled step line to finish. *Extension / fast-finisher* — break a two-action job into steps. *Reteach* — act the behavior out with bodies, naming each step, then map to blocks.

Exit check: Pair's sticky-note steps match the blocks in their working program. **At-home (optional):** Break "pour a bowl of cereal" into 5 little steps with a grown-up. ~5 min.

Lesson 4 — Basics: Say the Plan First (Picture Plan)

Objective (I can...): I can tell my plan in words or pictures first, then turn it into blocks. **CS concept:** Planning first / early algorithm planning — saying the plan in plain words is a first taste of **picking the big steps and hiding the little mess.** · **Builds on:** Lesson 3 (little steps).

AI Connection (2 min, read aloud): Grown-ups who make AI say its plan in plain words first — "if it sees a face, wave" — before any code. *Ask the group:* "Why say your plan out loud before you build it?"

Materials: LEGO CS & AI kit + device per pair, a picture "plan mat" per pair, a stack of blank "mission cards."

Model reuse: same Basics model from Lesson 2 — code plus one tiny brick tweak.

60-minute flow - Warm-up (10): "Secret mission" — each pair draws or says a 3-step plan on a mission card, then trades cards with another pair. Your card is the other pair's mission to build and run. Clear plans win the mission. - **Concept + demo (8):** Coach says 3 plain-word lines, then matches each to a picture block while kids watch — the plan is the bridge from idea to code. (*Try "guess what it does" with thumbs up/down before Run.*) - **Build/Code (25):** Pairs build the mission card they got: say or draw the plan on the mat, then code it step-by-step on the built model and run to check it matches. **Make-it-yours beat:** add or swap one brick to fit the mission's theme. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change one plan step (like a longer wait) and fix the matching block so plan and code still agree. - **Share + tidy-up (10):** Pairs point to their plan, then run to prove it matches. Model stays built.

Pair roles: Navigator owns the plan mat; Driver turns each line into a block. Swap at the wiggle break. (*Trio: third child checks each plan step got coded.*)

Differentiation: *Support* — a plan mat with picture slots to fill ("First . **Then** ."). *Extension / fast-finisher* — a 5-step plan. *Reteach* — code one plan step together, then let them do the next alone.

Exit check: Pair's plan steps line up one-to-one with their working blocks. **At-home (optional):** Draw a 4-picture plan for getting ready for bed. ~5 min.

Lesson 5 — Basics: Design Challenge — Build-a-Bot Parade

Objective (I can...): I can plan, build, and code a little machine that does steps I chose, in order. **CS**

concept: Putting order, little steps, and a plan together · **Builds on:** Lessons 1–4. *Also serves as flex/catch-up — if earlier lessons ran long, use today to finish and celebrate.*

AI Connection (2 min, read aloud): Making an AI toy is the same loop you'll use today: break it down → plan → build → try it. *Ask the group:* "Which of those steps do you think people redo the most?"

Materials: LEGO CS & AI kit + device per pair, plan mats, a simple picture challenge card. **Model already built** (Basics model from Lesson 2) — so the 25 minutes is plan + code + play, not build-from-scratch. Pairs may add small bricks only.

60-minute flow - Warm-up (10): Review song/carousel — fast call-and-response: "What's a program?" "Steps in order!" "What's breaking it down?" "Little steps!" - **Concept + demo (8):** Show the open challenge: make the bot do 3+ steps in an order you choose. Coach says a plan for one idea, then hands off. - **Build/Code (25):** Pairs plan and code a 3+ step program on the built model (small brick changes allowed). Coach cheers and nudges, doesn't solve. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add one "surprise" step (light, sound, or extra move) and add it to the plan. - **Share + tidy-up (10): Build-a-Bot parade** — bots march on their trays in a gallery walk; each pair names the one thing that helped most. **Coach photographs each build for the unit parent note.**

Pair roles: Co-plan, then Driver codes / Navigator points to the plan; swap at the wiggle break. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a 2-step target with a partial plan to finish. *Extension / fast-finisher* — require 5 steps with a light and a sound. *Reteach* — hand them their Lesson 4 plan as a starting template.

Exit check: Pair shows a working 3+ step program with a matching plan and can tell what it does. **Progress layer:** Send home the **Unit 1 parent note** ("your child can plan and code steps in order") with the **build photo**. **At-home (optional):** Draw a dream robot and label 3 things it does in order. ~5 min.

Lesson 6 — Hardware: Build the Machine & Start the Motor

Objective (I can...): I can build the hardware machine and use code to start and stop one motor. **CS**

concept: How hardware works — code controls a single motor (start/stop) · **Builds on:** Lesson 5 (we run ordered steps; now those steps drive a real part).

AI Connection (2 min, read aloud): Every smart robot has real parts — motors to move, sensors to notice. AI is the "thinking," but the motor is the "doing." *Ask the group:* "What jobs could a spinning motor do for a little robot?"

Materials: LEGO CS & AI kit + device per pair. **This is the unit's build lesson** — build the Hardware machine (hub + one motor) now and keep it assembled through Lesson 10.

60-minute flow - Warm-up (10): "Motors around us" — kids name things with motors (fan, toy car, blender) and spin their arms like a motor; on "stop!" they freeze. Start and stop, start and stop. - **Concept + demo (8):** Build is coming — first show the Start motor and Stop motor picture blocks. Run one so the motor spins, press Stop so it halts. Point: the code is bossing a real part. - **Build/Code (25):** Pairs **build the Hardware machine** (the unit's one build — hub plus one motor), then code it to start the motor, wait, and stop. Try it a few times. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make the motor start, stop, then start again — a start-stop-start pattern. Guess before running. - **Share + tidy-up (10):** Pairs show starting and stopping and point to the motor. **Leave the machine built.**

Pair roles: Driver builds/codes, Navigator points to the motor and says "start... stop." Swap at the wiggle break. (*Trio rotates build/code/test.*)

Differentiation: *Support* — a palette card showing the start and stop pictures. *Extension / fast-finisher* — add a wait so the motor runs for a set time, then stops. *Reteach* — press Start and Stop by hand, naming each, before putting the blocks in.

Exit check: Pair's code reliably starts and stops the motor on purpose. **At-home (optional):** Find 2 things at home with a motor; tell a grown-up how you'd turn each on and off. ~5 min.

Lesson 7 — Hardware: Fast, Slow, Forward, Back

Objective (I can...): I can make the motor move in different ways — faster, slower, forward, and back. **CS concept:** Investigating motor movements (speed, direction, and how long) · **Builds on:** Lesson 6 (start/stop a motor).

AI Connection (2 min, read aloud): A delivery robot speeds up on a clear path and slows down near people — it controls its motors carefully. *Ask the group:* "When should a robot go slow instead of fast?"

Materials: LEGO CS & AI kit + device per pair, a short "road" strip of tape on the table or floor. **Model reuse:** Hardware machine from Lesson 6 — code only.

60-minute flow - Warm-up (10): "Fast and slow" — kids march in place super fast, then slow-motion, then backward on "reverse!" Feel speed and direction with their bodies. - **Concept + demo (8):** Show the motor-speed picture and the direction picture. Run the motor fast, then slow, then the other way, so kids see the change. - **Build/Code (25):** Pairs explore motor movements: code the motor to go fast, then slow, then reverse, watching each time. On the "road" strip, they try to make the machine move a short way and stop near a line. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make the motor do "fast, then slow, then back" in one program; guess what the machine will do before running. - **Share + tidy-up (10):** Pairs show their favorite motor move and name it (fast / slow / back). Model stays built.

Pair roles: Driver codes the motor moves, Navigator calls out "faster, slower, back" and watches the result. Swap at the wiggle break. (*Trio rotates code/call/watch.*)

Differentiation: *Support* — change only speed (fast vs. slow). *Extension / fast-finisher* — make the machine travel to a line and stop. *Reteach* — change one setting at a time and run after each change so the difference is clear.

Exit check: Pair can make the motor change speed and direction on purpose. **At-home (optional):** Watch a fan or toy that speeds up and slows down; tell a grown-up what makes it change. ~5 min.

Lesson 8 — Hardware: Meet the Color Sensor

Objective (I can...): I can use the color sensor to tell what color it sees. **CS concept:** How a sensor works — the color sensor reads information from the world · **Builds on:** Lesson 7 (we know the motor; now meet the other part).

AI Connection (2 min, read aloud): A self-driving car's camera "sees" colors and signs; your color sensor "sees" colors too — that's how a machine takes in information. *Ask the group:* "Who does a self-driving car help? Could it ever worry someone?"

Materials: LEGO CS & AI kit + device per pair, a set of big color cards (red, green, blue, yellow). **Model reuse:** Hardware machine from Lesson 6; snap the color sensor on as a tiny change if it isn't already — no full rebuild.

60-minute flow - Warm-up (10): "Color hunt" — coach calls a color, kids touch something that color in the room, fast. Then: "How does a machine know a color if it has no eyes?" (It has a sensor.) - **Concept + demo (8):** Attach/point to the color sensor. Hold a color card to it and show the live color picture change on screen. Swap cards; the picture follows. - **Build/Code (25):** Pairs watch the live color reading as they hold up different cards, and code a simple "when I see red, light up red" style reaction so the machine shows what color the sensor sees. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Find a color the sensor gets confused by (a tricky in-between color); talk about why it's hard. - **Share + tidy-up (10):** Pairs show the sensor naming two colors correctly. Model stays built.

Pair roles: Driver codes and runs, Navigator holds color cards to the sensor and reads the on-screen color. Swap at the wiggle break. (*Trio rotates code/hold/read.*)

Differentiation: *Support* — use just two very different colors (red and blue). *Extension / fast-finisher* — make a different light or sound for three colors. *Reteach* — watch the live color picture while slowly moving one card in and out.

Exit check: Pair can show the color sensor correctly telling two different colors apart. **At-home (optional):** Find a machine at home that senses something (a light that turns on by itself); tell a grown-up what it notices. ~5 min.

Lesson 9 — Hardware: When the Sensor Talks to the Motor

Objective (I can...): I can make the color sensor and the motor work together — the color decides what the motor does. **CS concept:** How hardware parts interact — a color reading controls the motor · **Builds on:** Lesson 8 (color sensor) and Lesson 6 (motor).

AI Connection (2 min, read aloud): In a real robot, the "eyes" tell the "wheels" what to do — sensor talks to motor. AI robots do this all day. *Ask the group:* "If the sensor sees red, what would be a good thing for the motor to do?"

Materials: LEGO CS & AI kit + device per pair, color cards (red = stop, green = go). **Model reuse:** Hardware machine from Lesson 6 — code only.

60-minute flow - Warm-up (10): "Red light, green light" with bodies — green card, everyone walks; red card, freeze. The color is telling the "motors" (legs) what to do. - **Concept + demo (8):** Demo: when the sensor sees green, the motor spins; when it sees red, the motor stops. Hold up each card and show the motor obey. - **Build/Code (25):** Pairs code the sensor-to-motor link: green seen → motor go; red seen → motor stop. They test with real cards. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a third color that makes the motor go slow or backward. - **Share + tidy-up (10):** Pairs run their "traffic motor" with the cards and say which color does what. Model stays built.

Pair roles: Driver codes, Navigator works the color cards and watches the motor. Swap at the wiggle break. (*Trio rotates code/cards/watch.*)

Differentiation: *Support* — just two colors, go and stop. *Extension / fast-finisher* — three colors doing three different motor moves. *Reteach* — do green-go alone first, confirm it works, then add red-stop.

Exit check: Pair's motor changes what it does based on the color the sensor sees. **At-home (optional):** Watch a real traffic light; tell a grown-up what cars do at each color. ~5 min.

Lesson 10 — Hardware: Design Challenge — Color Signal Machine

Objective (I can...): I can design a machine that uses color to stand for a message, and the motor acts on it.

CS concept: Using color to represent information (a color means something) · **Builds on:** Lessons 6-9. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): Colors carry messages everywhere — red means stop, green means go. Computers use colors and signals to stand for information too. *Ask the group:* "What message could YOUR machine's colors send?"

Materials: LEGO CS & AI kit + device per pair, plan mats, color cards. **Model already built** (Hardware machine from Lesson 6) — the 25 min is plan + code + play, small changes only.

60-minute flow - Warm-up (10): Hardware review — pairs name the two parts (motor, color sensor) and do a quick action for each ("spin!" "I see red!"). (*Theme: Signal fair.*) - **Concept + demo (8):** Challenge: build a "color signal machine" where each color means something and the motor reacts (green = open the gate, red = close it, yellow = wiggle). Coach says a plan for one color, then hands off. - **Build/Code (25):** Pairs plan and code a machine where at least two colors each send a different message to the motor, on the built machine. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a third color-message, or add a light so the signal is easy to see. - **Share + tidy-up (10): Signal fair** — pairs show their color cards and the machine obeying each one. The audience guesses what each color means before the pair runs it. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator works the cards and points to the plan; swap at the wiggle break. (*Trio rotates plan/code/test.*)

Differentiation: *Support* — a two-color target with a partial plan. *Extension / fast-finisher* — three colors with three different motor jobs. *Reteach* — reuse the Lesson 9 program and change what each color does.

Exit check: Pair shows a working machine where colors stand for messages the motor acts on, with a matching plan. **Progress layer:** Send home the **Unit 2 parent note** ("your child knows the motor and color sensor and can make them work together") with the **build photo**. **At-home (optional):** Make up a color code for your room (green = come in, red = knock first); draw it. ~5 min.

Lesson 11 — Events: Build the Model & First Trigger

Objective (I can...): I can build the events model and make my program start when something happens. **CS concept:** Events (an action that causes something to happen — a trigger starts code) · **Builds on:** Lesson 10 (the machine reacts; now we name the thing that starts it).

AI Connection (2 min, read aloud): AI reacts to what it senses — a camera noticing your face is an "event" that starts something. *Ask the group:* "What could our robot notice that would make it start?"

Materials: LEGO CS & AI kit + device per pair. **This is the unit's build lesson** — build the Events model now and keep it assembled through Lesson 15.

60-minute flow - Warm-up (10): "What made it happen?" — coach calls a trigger (doorbell rings!) and kids do the response (run to the door!). Keep it loud and moving: knock → wave, clap → jump. Name it: the trigger is the EVENT. - **Concept + demo (8):** Show "start when I press Run" vs. "start when something happens." Demo a program that runs only when a button/tap event fires. - **Build/Code (25):** Pairs **build the Events model** (the unit's one build), then put a short sequence under an event picture (like a button press) so it acts only when triggered. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Swap in a different trigger; guess, then test, how the start changes. - **Share + tidy-up (10):** Pairs show their event-triggered program and name the trigger. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator points to the picture steps and names the trigger. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — a palette card showing the event blocks. *Extension / fast-finisher* — use two different triggers. *Reteach* — press the button with no event block (nothing happens), then add the event together.

Exit check: Pair's program runs only when the chosen event happens, not on its own. **At-home (optional):** Find 3 things at home that "start" something (light switch, doorbell). ~5 min.

Lesson 12 — Events: Red Light, Green Light

Objective (I can...): I can use a color event like a traffic signal to start and stop my machine. **CS concept:** A color reading as an event (traffic-signal style events) · **Builds on:** Lesson 11 (events as start signals) and Unit 2 (color sensor).

AI Connection (2 min, read aloud): A self-driving car treats a green light as an event that means "go" and a red light as "stop" — a color starts an action. *Ask the group:* "Why is it so important the car reacts to the right color?"

Materials: LEGO CS & AI kit + device per pair, red/green/yellow color cards. **Model reuse:** Events model from Lesson 11 — code only.

60-minute flow - Warm-up (10): "Traffic game" — green card, kids tiptoe forward; red card, freeze; yellow, wiggle in place. Fast changes for giggles. Each color is an event. - **Concept + demo (8):** Demo a color event: when the sensor sees green, the motor starts (go); when it sees red, it stops. The color is the trigger. - **Build/Code (25):** Pairs code color events — green seen → go, red seen → stop — so the machine behaves like a tiny traffic signal. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a yellow event that makes the machine slow down or beep as a warning. - **Share + tidy-up (10):** Pairs run their traffic machine with the cards and name each color event. Model stays built.

Pair roles: Driver codes the color events, Navigator works the cards and calls the color. Swap at the wiggle break. (*Trio rotates code/cards/watch.*)

Differentiation: *Support* — just green-go and red-stop. *Extension / fast-finisher* — add yellow as a third event. *Reteach* — set up green-go first, test, then add red-stop.

Exit check: Pair's machine starts and stops correctly from color events. **At-home (optional):** Watch a crossing signal; tell a grown-up what each color tells people to do. ~5 min.

Lesson 13 — Events: Make Your Own Triggers

Objective (I can...): I can create my own events in a program and choose what each one starts. **CS**

concept: Creating events in programs (choosing triggers and their actions) · **Builds on:** Lesson 12 (color events).

AI Connection (2 min, read aloud): People who build AI decide which events matter — "wake up when you hear my voice." They pick the triggers. *Ask the group:* "If you could pick one word to wake up a robot, what would it be?"

Materials: LEGO CS & AI kit + device per pair, color cards and (if the kit has it) a button/tap input. **Model reuse:** Events model from Lesson 11 — code only.

60-minute flow - Warm-up (10): "Pick the trigger" — coach offers choices ("should a clap or a stomp start the dance?"); kids vote and act it out. They're choosing events. - **Concept + demo (8):** Show how to drop an event picture and attach any action to it. Build one event live, then a second, different one. - **Build/Code (25):** Pairs create at least two of their own events (a press or tap → one action; a color → another action) and choose what each starts. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make one event start a two-step action (move then light) while another does something simple. - **Share + tidy-up (10):** Pairs show both of their events and say which trigger starts which action. Model stays built.

Pair roles: Each partner owns one event they create; swap at the wiggle break to try each other's. (*Trio: three events, one each.*)

Differentiation: *Support* — one event, with a twin example to copy. *Extension / fast-finisher* — add a third event. *Reteach* — build one event start-to-finish together, then mirror it for the second.

Exit check: Pair's program has two+ events they created, each starting a different action. **At-home**

(optional): Invent a secret trigger (a knock pattern) that means "come in"; teach it to a grown-up. ~5 min.

Lesson 14 — Events: The Machine Answers Back

Objective (I can...): I can make the hardware answer an event right away — the sensor or press makes the motor respond. **CS concept:** Hardware responding to events · **Builds on:** Lesson 13 (creating events).

AI Connection (2 min, read aloud): An always-on helper waits for an event and answers the moment it happens. *Ask the group:* "Who does an always-listening speaker help? Could it bother anyone at home?"

Materials: LEGO CS & AI kit + device per pair, color cards/trigger objects. **Model reuse:** Events model from Lesson 11 — code only.

60-minute flow - Warm-up (10): "Answer back" — coach gives a signal (raise a hand), kids instantly respond (stand up). The faster the better. Event in, response out. - **Concept + demo (8):** Demo an event that makes the motor respond right away — press the button, the motor spins; show the color sensor event making the motor move. Hardware is answering the event. - **Build/Code (25):** Pairs code events so the motor (and a light or sound) responds the instant a trigger happens — press → motor spins; color seen → motor reverses. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second event so the machine has two different hardware responses. - **Share + tidy-up (10):** Pairs trigger each event and show the motor answering. Model stays built.

Pair roles: Driver codes, Navigator triggers the event and watches the motor respond. Swap at the wiggle break. (*Trio rotates code/trigger/watch.*)

Differentiation: *Support* — one event → one motor response. *Extension / fast-finisher* — two events with two different motor responses. *Reteach* — press and watch nothing until the event block is added, then add it together.

Exit check: Pair's hardware (motor) responds correctly the moment the event happens. **At-home (optional):** Find something that answers right away when you do something (a light switch, a clap-on lamp); tell a grown-up the event and the response. ~5 min.

Lesson 15 — Events: Design Challenge — Press-to-Go Gadget

Objective (I can...): I can design a gadget that uses events to make the motor go, and I can program a motor with my chosen events. **CS concept:** Programming a motor with events (applying events to a purposeful build) · **Builds on:** Lessons 11-14. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): A smart doorbell reacts to events — motion AND a face it knows. *Ask the group:* "For YOUR gadget, should it need both triggers, or just one?"

Materials: LEGO CS & AI kit + device per pair, plan mats, color cards/trigger objects. **Model already built** (Events model from Lesson 11) — the 25 min is plan + code + play, small changes only.

60-minute flow - Warm-up (10): Rapid review — pairs name every event they used (press, tap, color, sensor) and act one out. (*Theme: Gadget fair.*) - **Concept + demo (8):** Challenge: build a "press-to-go gadget" that makes the motor do something fun when two different events happen. Coach says a plan for one event, then steps back. (*Try "the silly robot."*) - **Build/Code (25):** Pairs plan and code two events that each drive the motor (press → spin a spinner; color seen → open a gate) on the built model. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a third event or make one motor response a two-step move. - **Share + tidy-up (10): Gadget fair** — before a pair sets off their gadget, the audience guesses which events make the motor go, then the pair triggers them live to check. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator works the triggers and points to the plan; swap at the wiggle break. (*Trio rotates plan/code/test.*)

Differentiation: *Support* — a one-event target with a partial plan. *Extension / fast-finisher* — require a color event AND a press event. *Reteach* — reuse a Lesson 13-14 program as a starting template.

Exit check: Pair shows a working gadget where two+ events each make the motor go, with a matching plan.

Progress layer: Send home the **Unit 3 parent note** ("your child can make a model sense and react to events") with the **build photo**. **At-home (optional):** Invent a gadget for something at home; name the events that make it go. ~5 min.

Lesson 16 — Sequences: Build the Model & Run a Sequence

Objective (I can...): I can build the sequences model and run an ordered set of steps that finishes a job. **CS concept:** Sequences (ordered steps that work together to do a job) · **Builds on:** Lesson 15 (events start a program; now we grow the ordered steps after the start).

AI Connection (2 min, read aloud): A voice helper follows a sequence every time: wake → listen → think → answer. The order is always the same. *Ask the group:* "What would go wrong if it answered before it listened?"

Materials: LEGO CS & AI kit + device per pair, 4–5 big "step" picture cards per pair. **This is the unit's build lesson** — build the Sequences model now and keep it assembled through Lesson 20.

60-minute flow - Warm-up (10): "Morning march" — pairs put 5 "getting ready" picture cards in order (wake, dress, eat, shoes, go) and march through them. A sequence is steps in the right order. - **Concept + demo (8):** Show a 4–5 step program running top to bottom (light → motor spins → wait → beep → stop). Point to each step as it runs. The whole thing is a sequence. - **Build/Code (25):** Pairs **build the Sequences model** (the unit's one build), then code a 4–5 step sequence that finishes a little job, running it to check each step happens in order. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add one more step in the right spot; run and point to where it fits. - **Share + tidy-up (10):** Pairs run their sequence and count the steps out loud. **Leave the model built.**

Pair roles: Driver codes, Navigator points to the step cards and counts along. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — a 3-step sequence with the first step placed. *Extension / fast-finisher* — a 6-step sequence. *Reteach* — run one step at a time and name it before adding the next.

Exit check: Pair runs a 4+ step sequence in the correct order. **At-home (optional):** Draw a 5-picture sequence for making a snack, in order. ~5 min.

Lesson 17 — Sequences: Fix-It — The Steps Are Jumbled

Objective (I can...): I can find the step that's in the wrong place and fix the sequence so it works. **CS concept:** Debugging and developing a sequence (finding and fixing order mistakes) · **Builds on:** Lesson 16 (running a sequence).

AI Connection (2 min, read aloud): When AI acts weird, the people who build it hunt for the step that's out of order and fix it — just like us. *Ask the group:* "How could you tell WHICH step is the wrong one?"

Materials: LEGO CS & AI kit + device per pair, a printed "broken sequence" picture strip per pair. **Model reuse:** Sequences model from Lesson 16 — code only.

60-minute flow - Warm-up (10): "Silly order" — coach acts out a jumbled routine (shoes before socks) and kids shout the fix. Finding the mistake is called debugging. - **Concept + demo (8):** Show a short program whose steps are out of order so the job fails; think aloud, spot the wrong step, move it, and run again. (*Try "the silly robot."*) - **Build/Code (25):** Pairs get a "jumbled" program (a step in the wrong place), run it, see it fail, then **find and fix** the order so it works. Then they develop it: add one new step in the right spot. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make a sequence with a hidden order mistake and trade with another pair to fix each other's. - **Share + tidy-up (10):** Pairs show the "before" (broken) and "after" (fixed) and point to the step they moved. Model stays built.

Pair roles: Driver runs and fixes, Navigator watches for the wrong step and says where it goes. Swap at the wiggle break. (*Trio: third leads the hunt.*)

Differentiation: *Support* — only two steps are swapped. *Extension / fast-finisher* — fix a longer program with two mistakes. *Reteach* — run step by step and stop the second it looks wrong.

Exit check: Pair finds and fixes an order mistake so the sequence works, and can say what was wrong. **At-home (optional):** Mix up the steps of a bedtime routine on purpose; have a grown-up fix the order. ~5 min.

Lesson 18 — Sequences: A Sequence That Solves a Problem

Objective (I can...): I can build a sequence that gets a real little job done. **CS concept:** Programming a sequence to solve a problem · **Builds on:** Lesson 17 (fixing and developing sequences).

AI Connection (2 min, read aloud): A robot vacuum follows a sequence to clean a room — go, turn, go, turn — until the job is done. *Ask the group:* "What's a little job you'd love a robot to finish for you?"

Materials: LEGO CS & AI kit + device per pair, a simple "problem card" (e.g., move a block from here to there, or guard a spot). **Model reuse:** Sequences model from Lesson 16 — code only.

60-minute flow - Warm-up (10): "What's the job?" — coach names a little problem (the toy needs to get to the box); pairs say the steps out loud to solve it. - **Concept + demo (8):** Pick a problem card; coach decomposes it into an ordered sequence and codes it live, running to check it solves the job. - **Build/Code (25):** Pairs pick a problem card and code a sequence that solves it on the built model (move the block, deliver the mail, do the dance). They test and adjust until the job is done. *Wiggle break + swap at ~12 min.* -

Challenge (7): Solve the same problem a second way with different steps; which is simpler? - **Share + tidy-up (10):** Pairs show their machine solving the job and name the key step. Model stays built.

Pair roles: Co-plan; Driver codes / Navigator checks each step solves part of the job; swap at the wiggle break. (*Trio rotates plan/code/test.*)

Differentiation: *Support* — a problem card with the first two steps given. *Extension / fast-finisher* — a problem that needs 6+ steps. *Reteach* — act out solving the problem with bodies first, then map to blocks.

Exit check: Pair's sequence actually solves the problem on the card. **At-home (optional):** Pick a little problem at home (socks all over); write the steps to solve it. ~5 min.

Lesson 19 — Sequences: Draw the Sequence First

Objective (I can...): I can draw my sequence in pictures first (picture pseudocode), then build it. **CS**

concept: Sequences with simple picture pseudocode (plan in pictures, then code) · **Builds on:** Lesson 18 (sequences that solve problems).

AI Connection (2 min, read aloud): Engineers sketch an AI's steps as little pictures before they code — a picture plan. *Ask the group:* "Why might drawing your plan help before you build it?"

Materials: LEGO CS & AI kit + device per pair, a picture "storyboard" strip (4-5 empty boxes) per pair. **Model reuse:** Sequences model from Lesson 16 — code only.

60-minute flow - Warm-up (10): "Storyboard it" — pairs draw a 4-box picture plan for a tiny routine (make a sandwich) and point through it. Pictures can be a plan. - **Concept + demo (8):** Coach draws a 4-picture plan in the boxes, then matches each box to a picture block while kids watch — picture plan to code. - **Build/Code (25):** Pairs draw a 4-5 box picture plan for a sequence, then code it box-by-box on the built model and run to check code matches the plan. **Make-it-yours beat:** decorate the model to fit the plan. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change one picture in the plan and update the matching block so plan and code still agree. - **Share + tidy-up (10):** Pairs hold up their picture plan next to the running program and point box-to-block. Model stays built.

Pair roles: Navigator owns the storyboard; Driver turns each box into a block. Swap at the wiggle break. *(Trio: third checks each box got coded.)*

Differentiation: *Support* — a storyboard with 2 boxes pre-drawn. *Extension / fast-finisher* — a 6-box plan. *Reteach* — code one box together, then let them do the next alone.

Exit check: Pair's picture plan lines up box-to-block with their working sequence. **At-home (optional):** Draw a 4-box picture plan for feeding a pet or watering a plant. ~5 min.

Lesson 20 — Sequences: Design Challenge — Tell a Story in Order

Objective (I can...): I can program a sequence that tells a little story, in the right order. **CS concept:** Programming a sequence to tell a story (applying ordered steps for a purpose) · **Builds on:** Lessons 16–19. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): A storytelling app plays its scenes in order, one after another, to make sense. *Ask the group:* "What happens to a story if the ending plays first?"

Materials: LEGO CS & AI kit + device per pair, picture storyboards, story props. **Model already built** (Sequences model from Lesson 16) — the 25 min is plan + code + play, small changes only.

60-minute flow - Warm-up (10): Sequences review — pairs put a 4-scene story (beginning, middle, surprise, end) in order and tell it fast. (*Theme: Story theater.*) - **Concept + demo (8):** Challenge: code a sequence that acts out a tiny story with the machine (lights, motor moves, sounds) in order. Coach storyboards one idea, then hands off. (*Try "show a friend."*) - **Build/Code (25):** Pairs storyboard then code a 4+ step story sequence on the built model, running to make sure the scenes play in order. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add one surprise scene in the right spot and update the storyboard. - **Share + tidy-up (10): Story theater** — pairs perform their machine's story in order while the group watches. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator narrates the story beats and points to the storyboard; swap at the wiggle break. (*Trio rotates plan/code/perform.*)

Differentiation: *Support* — a 3-scene story with a partial storyboard. *Extension / fast-finisher* — a 6-scene story with a light, a sound, and a motor move. *Reteach* — reuse their Lesson 19 picture plan as a starting story.

Exit check: Pair's machine tells a 4+ scene story with the scenes in the right order, matching a storyboard.

Progress layer: Send home the **Unit 4 parent note** ("your child can build, fix, and tell a story with an ordered program") with the **build photo**. **At-home (optional):** Storyboard a 4-scene story and act it out for your family in order. ~5 min.

Lesson 21 — Loops: Build the Model & Do It Again

Objective (I can...): I can build the loops model and use a repeat block so I don't copy the same block many times. **CS concept:** Loops / repeating actions (repeat N times). A loop is **hiding the mess**: one "repeat 5x" picture stands in for five copied blocks. · **Builds on:** Lesson 20 (an ordered story; now we repeat a chunk of it).

AI Connection (2 min, read aloud): AI gets better by repeating — it sees examples over and over to learn.

Ask the group: "Why would doing it many times help a computer learn?"

Materials: LEGO CS & AI kit + device per pair. **This is the unit's build lesson** — build the Loops model now and keep it assembled through Lesson 25.

60-minute flow - Warm-up (10): "Do it again" — clap 10 times counting together, then find a shorter way to say it: "repeat 10 times!" Sing a repeating song (a 2-line chorus over and over) and name the repeat. -

Concept + demo (8): Show 4 copied "move" blocks, then replace them with one move inside a "repeat 4" picture; run both so kids see they match. - **Build/Code (25):** Pairs **build the Loops model** (the unit's one build), then code a repeated action with a repeat picture (spin, pause, repeat 5x). *Wiggle break + swap at ~12 min.* -

Challenge (7): Change the repeat number; guess how much longer it runs before you press Run. -

Share + tidy-up (10): Pairs show their loop and say the count out loud. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator counts the repeats out loud. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — a palette card highlighting the repeat picture. *Extension / fast-finisher* — put two blocks inside the loop. *Reteach* — count the repeats out loud together on the first run.

Exit check: Pair swaps copied blocks for a working repeat loop and can say the count. **At-home (optional):** Find something you do over and over; say it as "repeat ___ times." ~5 min.

Lesson 22 — Loops: Loops with Moves & Sounds (+ Fix-It)

Objective (I can...): I can develop a program that puts a few actions inside one loop so a whole pattern repeats. **CS concept:** Developing a program with loops (a loop body with more than one action) · **Builds on:** Lesson 21 (single-action repeat).

AI Connection (2 min, read aloud): Each training "round" repeats the same steps on new examples until the AI gets better. *Ask the group:* "What might get a little better each time the loop runs again?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Loops model from Lesson 21 — code only.

60-minute flow - Warm-up (10): Pattern clap — coach does "clap-stomp-clap-stomp"; kids find the chunk that repeats and do it as a group. That chunk goes in the loop. - **Concept + demo (8):** Build a loop with move + sound + wait; run it and point out the whole chunk repeats. - **Build/Code (25):** Pairs develop a 3-action repeating pattern inside one loop (spin → beep → wait). **Fix-It mini-task (3 min):** hand each pair a "silly" program where a block sits OUTSIDE the loop — they find it and drag it in so the whole pattern repeats. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a light or a backward move to the loop; run it again. - **Share + tidy-up (10):** Pairs show the pattern and name the actions inside the loop. Model stays built.

Pair roles: Driver codes the loop, Navigator names the actions to include. Swap at the wiggle break. (*Trio: third leads the Fix-It hunt.*)

Differentiation: *Support* — a 2-action loop with a twin example. *Extension / fast-finisher* — a 4-action loop. *Reteach* — build the actions outside a loop first, confirm they work, then drag them inside together.

Exit check: Pair runs a loop whose body has 3+ actions that repeat together. **At-home (optional):** Make up a 3-move dance; repeat it 4 times for a grown-up. ~5 min.

Lesson 23 — Loops: Repeat and Keep Count

Objective (I can...): I can use a repeat loop and keep a simple count of how many times it ran. **CS concept:** Repeat loops + collecting and presenting simple data (counting the repeats) · **Builds on:** Lesson 22 (loop bodies).

AI Connection (2 min, read aloud): AI keeps a tally as it practices — how many rounds it has done, how many it got right. *Ask the group:* "What would a robot count to know it practiced enough?"

Materials: LEGO CS & AI kit + device per pair, a big tally sheet or counting mat per pair. **Model reuse:** Loops model from Lesson 21 — code only.

60-minute flow - Warm-up (10): Tally clap — coach claps in a loop; kids make a tally mark each time and count the total. We can collect data on how many times something happened. - **Concept + demo (8):** Run a repeat-5 loop and show a counter on screen (or a light that flashes once per repeat); the number IS our data. Point to how the count grows. - **Build/Code (25):** Pairs code a loop that repeats an action and shows the count (screen number or flashes), then record the count on their tally sheet. Run with two different repeat numbers and write both down. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make a little bar of blocks (one brick per repeat) to "show" the count — a first picture graph. - **Share + tidy-up (10):** Pairs show their loop, read the count aloud, and hold up their brick-bar of the data. Model stays built.

Pair roles: Driver codes and runs the loop, Navigator tallies and reads the count. Swap at the wiggle break. *(Trio: third builds the brick-bar.)*

Differentiation: *Support* — count just one small loop. *Extension / fast-finisher* — compare counts from two loops and say which is bigger. *Reteach* — count each repeat out loud together and mark it as it happens.

Exit check: Pair runs a repeat loop and collects and shows a simple count of the repeats. **At-home (optional):** Count how many times you can hop in one loop of a song; write the number. ~5 min.

Lesson 24 — Loops: Make Your Own Loops

Objective (I can...): I can create my own loop in a program and choose what repeats and how many times.

CS concept: Creating loops in a program · **Builds on:** Lesson 23 (repeat and count).

AI Connection (2 min, read aloud): The people who build AI set up its practice loops — what to repeat and how many rounds. *Ask the group:* "If you could set how many times a robot practices, what number would you pick?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Loops model from Lesson 21 — code only.

60-minute flow - Warm-up (10): "Your repeat" — each kid picks a move and a number ("jump, 3 times!"); the group does it. They're inventing loops. - **Concept + demo (8):** Show starting from an empty program: drop a repeat picture, choose the body, choose the count. Build one loop from scratch live. - **Build/Code (25):** Pairs create their own loop from scratch — they pick the actions for the body and the repeat number, then run and adjust. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make a second, different loop in the same program and run them one after the other. - **Share + tidy-up (10):** Pairs show the loop they made and say what repeats and how many times. Model stays built.

Pair roles: Each partner creates one loop; swap at the wiggle break to try each other's. (*Trio: three loops, one each.*)

Differentiation: *Support* — choose the count for a loop body that's already there. *Extension / fast-finisher* — create two loops with different bodies. *Reteach* — build one simple loop together, then let them make the next alone.

Exit check: Pair creates a working loop from scratch and can say what repeats and how many times. **At-home (optional):** Invent a chore loop ("fold a shirt, repeat 5 times"); do it with a grown-up. ~5 min.

Lesson 25 — Loops: Design Challenge — Pattern Party

Objective (I can...): I can design a machine that repeats a pattern I made up, using loops. **CS concept:** Using loops for a creative build · **Builds on:** Lessons 21–24. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): Repeating is how AI practices — lots and lots of loops to get one skill right. *Ask the group:* "Would your machine get better if it practiced its pattern more, like AI does?"

Materials: LEGO CS & AI kit + device per pair (with camera), plan mats, a big screen for the coach. **Model already built** (Loops model from Lesson 21) — build time is plan + code + play, small changes only.

60-minute flow *(As the flex lesson, today's clock trims Build/Code to ~18 to make room for a short AI peek. If a pair needs the full build time, run the AI peek as a whole-group demo instead.)* - **Warm-up (10):** Loop review — pairs sort example programs into "repeat N" and "loop with many actions," and say which is simpler than copying blocks. *(Theme: Pattern party — everyone runs at once at the end.)* - **Concept + demo (8):**

Challenge: build a "pattern machine" (dancer, drummer, light show) using at least one loop you made. Coach says a plan with a loop, then hands off. *(Try "show a friend.")* - **Build/Code (~18):** Pairs plan and code a repeating pattern that must use a loop on the built model. *Swap at the halfway mark.* - **Peek at a real AI — watch it recognize (10):** The **coach** opens the kit's AI pose blocks (or Google Teachable Machine on the coach's screen) and strikes poses — arms up, arms out, hands on head — so the whole group watches the on-screen label name each pose and calls it out. **This is a watch-and-feed demo — the kids are exploring how a computer recognizes poses, not building their own AI.** *This previews Unit 6.* **Privacy note (say it aloud):** the camera runs on the device, with no logins, and pictures are not saved. - **Share + tidy-up (10):** Pattern party — pairs run their pattern together like a band, and each points to the loop in their program. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator counts the loop; swap at the halfway mark. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a single-repeat target with a partial plan. *Extension / fast-finisher* — add a second loop plus a trigger. *Reteach* — reuse a Lesson 22–24 loop as a template.

Exit check: Pair shows a working pattern that uses a loop, and can say why one repeat block is easier than copying the blocks many times (**hiding the mess**). **Progress layer:** Send home the **Unit 5 parent note** ("your child can make a machine repeat a pattern and keep count with loops") with the **build photo**. **At-home (optional):** Make up a clap pattern; say how many times each part repeats. ~5 min.

Lesson 26 — AI & Data: Collect the Data (Build the Model)

Objective (I can...): I can collect simple data with my machine and describe what it shows. **CS concept:** Collecting, describing, and representing data · **Builds on:** Lesson 25 (we counted repeats; now we gather and describe data). *Honest framing: today kids collect and sort data themselves and watch a computer recognize — no AI is trained by the child.*

AI Connection (2 min, read aloud): AI starts with data — lots of examples collected and sorted. Before a computer can "know" a cat, people collect cat pictures. *Ask the group:* "If we wanted to teach a computer about colors, what data would we collect?"

Materials: LEGO CS & AI kit + device per pair, color cards, a big tally/graph mat per pair. **This is the AI unit's build lesson** — build the AI & Data model now and keep it assembled through Lesson 30.

60-minute flow - Warm-up (10): "Count the room" — pairs collect quick data (how many kids wore red? blue?) with tally marks, then say which group was biggest. That's collecting and describing data. - **Concept + demo (8):** The **coach** shows the color sensor reading colors and the device collecting the data (how many reds, how many greens). Describe it together: "We saw more green than red." On-device, saves nothing. - **Build/Code (25):** Pairs **build the AI & Data model** per the Canvas picture steps. Then they use the color sensor to collect data — show it several color cards and tally how many of each the machine reads — and describe the result on their graph mat (which color came up most?). *Wiggle break + swap at ~12 min.* - **Challenge (7):** Build a brick bar-graph (one brick per reading) so the data is easy to see. - **Share + tidy-up (10):** Pairs show their tally and brick graph and describe what the data shows. **Leave the model built.**

Pair roles: Driver builds and runs the sensor, Navigator tallies and describes the data. Swap at the wiggle break. (*Trio rotates build/collect/tally.*)

Differentiation: *Support* — collect just two colors. *Extension / fast-finisher* — collect three colors and say the order from most to least. *Reteach* — read one card at a time and mark the tally together.

Exit check: Pair collects simple data and can describe what it shows (which group had the most). **At-home (optional):** Collect data at home: count forks vs. spoons in a drawer; tell a grown-up which there are more of. ~5 min.

Lesson 27 — AI & Data: A Program That Shows Our Data

Objective (I can...): I can change a program so it collects data and shows it on screen. **CS concept:** Modifying a program that collects and visualizes data · **Builds on:** Lesson 26 (collecting and describing data).

AI Connection (2 min, read aloud): AI people make charts of their data to see it clearly — a picture of the numbers helps everyone understand. *Ask the group:* "Is it easier to understand a big list of numbers, or a picture of them?"

Materials: LEGO CS & AI kit + device per pair, the AI & Data model (from L26), color cards. **Model reuse:** AI & Data model from Lesson 26 — mostly code.

60-minute flow - Warm-up (10): "Make it a picture" — pairs turn yesterday's tally into a quick picture (stack of stickers or blocks per color). A picture of data is easier to read. - **Concept + demo (8):** Show a starter program that collects color readings and shows a simple on-screen count or bar. Change one thing (add a color, change the display) and run so kids see the chart update. (*Try "guess what it does."*) - **Build/Code (25):** Pairs take the starter data program and **modify** it — add a color to collect, change what the screen shows, or add a light that flashes for the biggest group — then run it to collect and show their data. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Collect two rounds of data and compare the two charts: did the biggest group change? - **Share + tidy-up (10):** Pairs show their program collecting data and the on-screen picture of it, and read the chart aloud. Model stays built.

Pair roles: Driver modifies the program, Navigator feeds color cards and reads the chart. Swap at the wiggle break. (*Trio: third records the two rounds.*)

Differentiation: *Support* — change just one thing in the starter program. *Extension / fast-finisher* — add a third category and a label. *Reteach* — change one block, run, and see the chart move before changing the next.

Exit check: Pair modifies a program so it collects data and shows it on screen, and can read the chart. **At-home (optional):** Draw a simple bar chart of something you counted (books on a shelf by color). ~5 min.

Lesson 28 — AI & Data: Meet the AI Pose Blocks

Objective (I can...): I can use a ready-made AI pose block that already knows how to spot body poses, and make my model react to a pose. **CS concept:** Using programs that use pretrained AI pose-recognition blocks.

Builds on: Lesson 27 (data) and Unit 3 events (a recognized pose is a trigger). *Honest framing: kids use ready-made AI pose blocks that are already trained — they do not train their own AI. Privacy: on-device camera, no logins, pictures never saved.*

AI Connection (2 min, read aloud): These pose blocks already learned, from lots of examples, how to tell "arms up" from "arms out." Today YOU use what the AI already knows — you don't have to teach it. *Ask the group:* "Who do you think showed these blocks all the example poses so they could learn?"

Materials: LEGO CS & AI kit + device per pair with camera, open floor space to pose. **Model reuse:** AI & Data model from Lesson 26 — code only.

60-minute flow - Warm-up (10): "Freeze pose" — coach calls a pose (arms up! star shape! hands on head!) and kids strike it and freeze. These are the kinds of poses the AI block can spot. - **Concept + demo (8):**

Demo the pretrained pose block: strike a pose in front of the camera and show the on-screen label name it. Say it plainly: **the block already learned this; we're using it, not teaching it.** On-device, saves nothing.

- **Build/Code (25):** Pairs use the pretrained pose block in a program so that when the camera recognizes a pose (arms up), the model reacts (motor spins or light on). They take turns posing and watching the model answer. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second pose that makes a different thing happen. - **Share + tidy-up (10):** Pairs strike their poses and show the model reacting, and say "the AI block already knew this pose." Model stays built.

Pair roles: Driver codes, Navigator strikes poses to the camera and reads the label. Swap at the wiggle break. *(Trio rotates code/pose/watch.)*

Differentiation: *Support* — one pose → one reaction. *Extension / fast-finisher* — two poses → two different reactions. *Reteach* — watch the on-screen label change as you pose before wiring the reaction.

Exit check: Pair's model reacts correctly when the pretrained pose block recognizes a pose. **At-home (optional):** Name 3 ways phones or tablets "recognize" things (faces, voices, poses). ~5 min.

Lesson 29 — AI & Data: Change the Pose Program

Objective (I can...): I can change a pose program to do what I want, and tell one reason the AI might guess wrong. **CS concept:** Modifying programs that use pretrained AI pose blocks; example quality and fairness ·

Builds on: Lesson 28 (using pose blocks). *Honest framing: kids modify a program that uses ready-made pose blocks and watch the coach's demo learn from examples — they do not train their own AI. Privacy: on-device camera, no logins, pictures never saved.*

AI Connection (2 min, read aloud): This is why we said "more tries help" back in loops — a computer learns poses from lots of different examples, and it can still guess wrong. *Ask the group:* "What kind of example would help a computer that keeps mixing up two poses?"

Materials: LEGO CS & AI kit + device per pair with camera, a big screen for the coach's Teachable Machine demo. **Model reuse:** AI & Data model from Lesson 26 — code only.

60-minute flow - Warm-up (10): "Tricky twins" — coach shows two poses that look alike; ask how you'd tell them apart for sure. Connect to clearer, more varied examples. - **Concept + demo (10):** On the **coach's** screen, run Google Teachable Machine: kids take turns showing example poses to the camera, the coach captures them, and the group watches the demo learn to tell poses apart — then sees it guess wrong when a pose only had one kind of example. **Say it plainly: the coach's computer is learning from the examples WE show it; the kids are exploring how it learns, not training their own AI.** On-device, no logins, pictures not saved. - **Build/Code (22):** Back at their own model, pairs **modify** their pose program from L28 — add or change a pose reaction, swap which pose does what — and test where it still guesses wrong. They keep a tally of "how many examples per pose" on the plan mat and ask "are our poses even?" *Wiggle break + swap at ~12 min.* - **Challenge (7):** Try a pose the block mixes up; talk about why it's tricky. - **Share + discussion (10):** Pairs report what they changed, then the group discusses **why** a computer still guesses wrong sometimes: maybe it only saw one kind of example, or one pose had way more examples than the other. Name one better example that would help, and talk about being fair — did the examples include all kinds of kids and poses? Model stays built.

Pair roles: Driver modifies the program, Navigator poses and reports right/wrong. Swap at the wiggle break. *(Trio: third keeps the example tally.)*

Differentiation: *Support* — change just one pose reaction. *Extension / fast-finisher* — add a third pose and balance the examples in the coach demo. *Reteach* — add one example at a time on the coach demo so the improvement is visible each time.

Exit check: Pair modifies a pose program so it does what they want, and can name one reason a computer might guess wrong and one way better examples would help. **At-home (optional):** Explain to a family member why a computer needs lots of different examples to learn poses well. ~5 min.

Lesson 30 — AI & Data: Design Challenge — AI Pose Story Circle

Objective (I can...): I can program events with AI pose blocks so a pose makes my build do something in a little story. **CS concept:** Programming events with AI pose blocks (applying pose recognition with events and data) · **Builds on:** Lessons 26–29 and the whole course. *Honest framing: kids use ready-made pose blocks — they do not train their own AI. Privacy: on-device camera, no logins, pictures never saved. Also serves as flex/catch-up — use today to finish and polish anything from the AI unit.*

AI Connection (2 min, read aloud): Today your build senses, collects data, and reacts to a pose that a ready-made AI block recognizes — every unit of the year working together. *Ask the group:* "Which idea from this year did your story use the most?"

Materials: LEGO CS & AI kit + device per pair with camera, plan mats, story props. **Model already built** (AI & Data model from Lesson 26) — the 25 min is plan + code + play, not build-from-scratch.

60-minute flow - Warm-up (10): Story spark — pairs pitch a one-sentence story where striking a pose changes what happens next. (*Theme: AI pose show.*) - **Concept + demo (8):** Challenge: program events with the pose blocks so a recognized pose drives the model and the story (arms up → the rocket launches). Coach shows a short plan connecting a pose to a story beat. Restate privacy points. - **Build/Code (25):** Pairs plan and code a little interactive story where at least one pose event drives the model, plus (if they can) a count or a loop, on the built model. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second pose that leads to a different ending. - **Share + tidy-up (10): AI pose show** — each pair performs their pose story while the coach records a short video clip for the parent note. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator directs the story and strikes the poses; swap at the wiggle break. (*Trio rotates plan/code/perform.*)

Differentiation: *Support* — one pose, one story beat, partial plan. *Extension / fast-finisher* — two poses with two endings and a count. *Reteach* — reuse their Lesson 28 program and add just one story reaction.

Exit check: Pair shows a working pose story where a pose event makes the build do something, with a matching plan. **Progress layer:** Send home the **Unit 6 (course finale) parent note** ("your child can collect data and use ready-made AI pose blocks to build a machine that senses, repeats, and responds — and can explain how a computer learns to recognize things from examples") with the **build photo** — the year's capstone. *Keep the framing honest: your child explored how AI works and used AI pose blocks; training a classifier of their own is next, in the 3–5 band.* **At-home (optional):** Draw a 3-picture pose story; mark which pose changes the ending. ~5 min.

Standards Crosswalk

Aligned to the **CSTA K-12 CS Standards** — primarily **Algorithms & Programming (1A-AP)**, with connections to **Data & Analysis (1A-DA)** and **Impacts of Computing (1A-IC)**. These are the Level 1A (grades K-2) codes, mapped to the six real units. This is a targeted map, not a claim of full coverage. *Maps to the Virginia CS Standards of Learning, grades K-2 (confirm exact SOL numbers).*

Unit (Lessons)	Skills taught	CSTA 1A codes targeted
1 • Basics (1-5)	Collaboration routines (Driver/Navigator), modeling everyday processes as algorithms, breaking a job into little steps, saying the plan first	1A-AP-08 (model daily processes as precise step-by-step algorithms); 1A-AP-10 (develop programs with sequences); 1A-AP-11 (decompose steps into a precise sequence)
2 • Hardware (6-10)	How the motor works (start/stop, speed, direction), how the color sensor reads, motor and sensor working together, using color to stand for information	1A-AP-10 (programs that control hardware); 1A-AP-09 (use color/symbols to represent information); 1A-AP-14 (test and fix motor behavior)
3 • Events (11-15)	Events that cause something to happen, traffic-signal color events, creating events, hardware responding to events, programming a motor with events	1A-AP-10 (events and control in programs); 1A-AP-08 (precise step-by-step behavior); 1A-AP-11 (decompose a triggered behavior)
4 • Sequences (16-20)	Ordered steps, debugging/developing a sequence, a sequence that solves a problem, picture pseudocode, a story in order	1A-AP-11 (order steps into a precise sequence); 1A-AP-10 (develop sequenced programs); 1A-AP-12 (develop plans that describe a program's sequence); 1A-AP-14 (debug)
5 • Loops (21-25)	Repeat loops, developing a loop body, repeating + collecting a simple count, creating loops, pattern builds	1A-AP-10 (loops/repetition in programs); 1A-AP-14 (debug/Fix-It); 1A-DA-06 (collect and present simple data — the repeat count)
6 • AI & Data (26-30)	Collecting/describing/representing data, modifying a program that collects & visualizes data, using pretrained AI pose blocks, modifying pose programs, programming events with pose blocks, why computers guess wrong (fairness)	1A-DA-05 (store, copy, and sort information to make it useful); 1A-DA-06 (collect and present data — tallies and simple charts); 1A-IC-16 (how people use computing / being fair); 1A-AP-10
Pair programming (every lesson)	Driver/Navigator roles, swapping, taking turns, following and giving instructions	1A-AP-12 (develop/follow plans together); collaboration across all lessons

Cross-cutting: Fix-It and debug mini-tasks (L10 catch-ups, L17, L19, L22) reinforce **1A-AP-14**; the "two ways to do it" and design-challenge share-outs reinforce **1A-AP-12**; the AI-impacts prompts (L8 self-driving cars, L14 always-listening speakers, and the L29 fairness discussion) reinforce **1A-IC-16**; the data-collection and tally work in Units 5-6 reinforces **1A-DA-05 / 1A-DA-06**.

Honesty note for coaches and families: Throughout Unit 6, K-2 students **explore how AI works** — they collect and show simple data, feed a coach-run Teachable Machine demo examples, and use **ready-made (pretrained) AI pose-recognition blocks** to drive a build. They **do not build or train their own AI model**. Training a custom classifier is a grade 3-5 skill and lives in the Builders and Engineers bands. Keep that promise accurate on every flyer, parent note, and share-out.